

ATS TRACKER IMPLEMENTATION

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional requirements and system features of the Applicant Tracking System (ATS). The system is designed to streamline and automate the end-to-end recruitment lifecycle including job requisition, candidate sourcing, evaluation, interview management, offer rollout, onboarding initiation, and recruitment analytics.

1.2 Scope

The ATS will serve as a centralized recruitment platform for:

- Administrators
- Recruiters
- Hiring Managers
- Vendors
- Employees (Referral users)
- Candidates

It will provide role-based access and workflow-driven hiring operations.

2. System Overview

The system enables:

- Job creation and approval
- Candidate sourcing from multiple channels
- Structured hiring pipeline management

- Offer generation and acceptance tracking
 - Vendor and referral management
 - Recruitment analytics and KPI monitoring
 - Process automation
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3. User Roles & Responsibilities

3.1 Administrator

- Configure system settings
- Manage users and roles
- Define workflows and approval processes
- Customize templates, modules, and layouts
- Manage integrations and compliance settings

3.2 Recruiter

- Create and manage job openings
- Source and manage candidates
- Schedule interviews
- Generate and send offer letters
- Track recruitment pipeline

3.3 Hiring Manager

- Approve job requisitions
- Review candidates
- Provide interview feedback
- Approve offers

3.4 Vendor

- View shared job openings
- Submit candidates
- Track submission status

3.5 Employee (Referral User)

- Refer candidates
- Track referral progress

3.6 Candidate

- Register and manage profile
- Apply for jobs
- Track application status

4. Functional Requirements

4.1 Dashboard & My Actions

- Personalized user dashboard
- Pending approvals
- Scheduled interviews
- Follow-ups and tasks
- Recruitment KPI widgets

4.2 Job Opening Management

- Create and manage job requisitions
- Define job attributes:
 - Title

- Department
 - Hiring manager
 - Number of positions
 - Employment type
 - Target dates
 - Job status tracking
 - Recruitment stage pipeline visualization
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4.3 Candidate Management

- Centralized candidate repository
- Candidate profile with:
 - Personal details
 - Contact details
 - Experience
 - Skill set
 - Resume
 - Address
- Candidate source tracking
- Candidate status lifecycle:
 - Associated
 - Screening
 - Evaluation
 - To-be-offered
 - Offered

- Hired
 - Rejected
 - Duplicate detection and record merge
 - Notes, attachments, and activity timeline
 - Ratings and structured feedback
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4.4 Interview Management

- Interview scheduling
 - Multi-stage interview tracking
 - Interview feedback capture
 - Interview notifications
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4.5 Offer Management

- Offer letter generation using templates
- Define:
 - Compensation
 - Employment type
 - Joining date
 - Offer expiry date
- Offer status tracking:
 - To-be-offered
 - Offered
 - Accepted
 - Declined

- Withdrawn
 - Digital signature integration
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4.6 Approval Workflow Management

- Multi-level approval configuration
 - Rule-based approval routing
 - Automated notifications
 - Post-approval task automation
 - Field updates based on approval outcome
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4.7 Candidate Portal

- Candidate self-registration
 - Profile completion tracking
 - Resume upload
 - Job search and application
 - Applied job tracking
 - Portal theme customization
 - Custom registration fields
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4.8 Career Site Management

- Public job listing
- Job detail pages
- Online application forms
- Custom career pages

- Website integration
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4.9 Vendor Management

- Vendor database
 - Vendor login access
 - Share job openings with vendors
 - Vendor candidate submission
 - Vendor performance tracking
 - Vendor activity dashboard
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4.10 Employee Referral Management

- Refer a candidate for a job
 - Capture referral relationship details
 - Referral history tracking
 - Referral status monitoring
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4.11 Resume Management

- Resume inbox
 - Resume parsing and field mapping
 - Bulk candidate import
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4.12 Reports & Analytics

KPI Metrics:

- Time-to-Hire

- Time-to-Fill
- Age of Job
- Offer Acceptance Rate

Report Capabilities:

- Recruiter performance reports
 - Department-wise hiring analysis
 - Pipeline metrics
 - Source effectiveness
 - Scheduled report generation
 - Export and email reports
 - Graphical visualization
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4.13 Workflow Automation

- Recruitment workflow configuration
 - Assignment rules for auto allocation
 - Email auto-response rules
 - Blueprint stage control
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4.14 Communication Management

- Email templates
 - Bulk email (mail merge)
 - Campaign management
 - Activity timeline tracking
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4.15 Integrations

- E-signature platforms
 - Background verification tools
 - Collaboration tools
 - Email services
 - Job boards
 - Telephony APIs
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4.16 User & Security Management

- Role-based access control
 - Territory management
 - Data privacy controls
 - Audit logs
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4.17 System Customization

- Custom fields and modules
 - Layout customization
 - Template configuration
 - Branding and theme management
 - Custom views (Classic / Customized)
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5. End-to-End Recruitment Workflow

1. Job requisition creation
2. Approval process

3. Job publishing (career site / vendors)
 4. Candidate application / submission
 5. Screening and evaluation
 6. Interview scheduling and feedback
 7. Offer generation
 8. Offer acceptance
 9. Hiring closure
 10. Recruitment analytics
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6. Non-Functional Requirements

6.1 Security

- Role-based authorization
- Secure data access
- Audit trail

6.2 Performance

- Fast candidate search and filtering
- Real-time dashboard updates

6.3 Scalability

- Support for large candidate databases
- Multi-department hiring

6.4 Usability

- Role-based dashboards
- Configurable views
- Intuitive navigation

6.5 Availability

- Web-based access
 - Multi-user concurrent access
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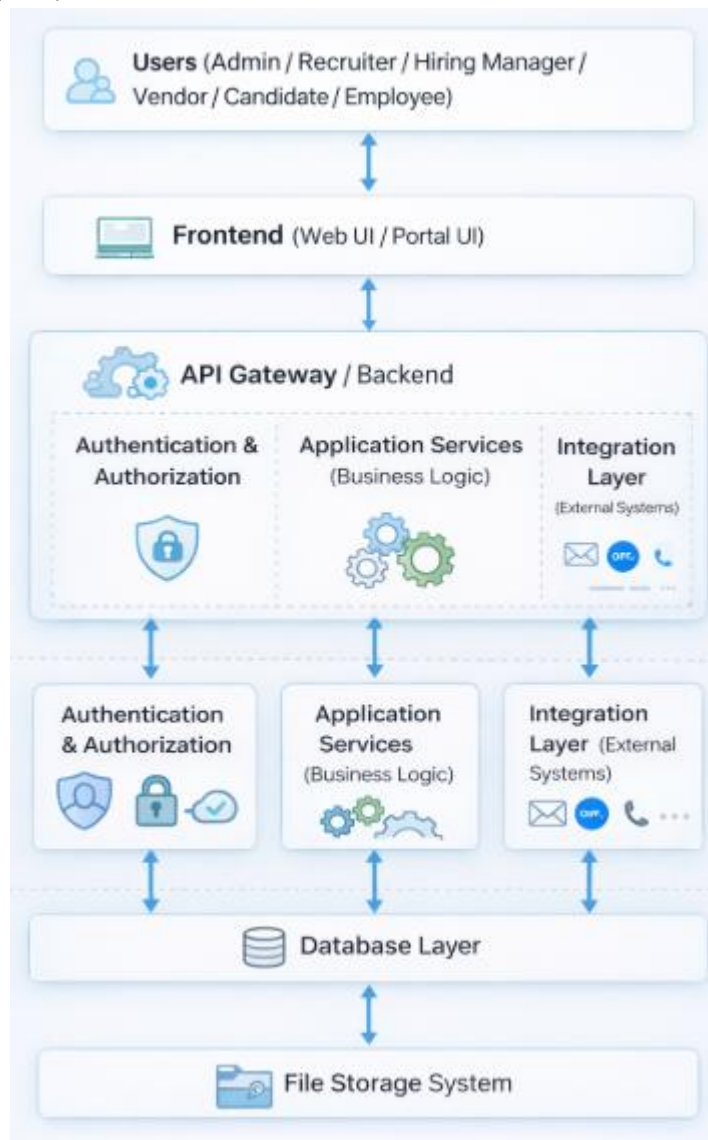
7. Key Benefits

- Reduced hiring cycle time
- Centralized recruitment data
- Improved collaboration
- Data-driven hiring decisions
- Automated workflow management
- Enhanced candidate experience

APPLICANT TRACKING SYSTEM – SYSTEM ARCHITECTURE

1. High-Level Architecture

Users (Admin / Recruiter / Hiring Manager / Vendor / Candidate / Employee)



2. Architecture Layers

2.1 Presentation Layer (Frontend)

Interfaces:

- Recruiter Web Dashboard
- Hiring Manager Interface
- Admin Setup Console
- Candidate Portal
- Vendor Portal
- Employee Referral Portal
- Career Site (Public)

Responsibilities:

- Role-based UI rendering
- Dashboard visualization
- Form handling
- Pipeline drag-and-drop
- Report charts

Recommended Tech:

- React.js / Angular / Vue.js
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2.2 API Gateway / Backend Layer

Responsibilities:

- Route requests
- Handle authentication

- Load balancing
- API security
- Request validation

Recommended Tech:

- Node.js (Express / NestJS)
OR
 - Java Spring Boot
OR
 - .NET Core
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2.3 Service Layer (Core Business Modules)

Each ATS function runs as a **separate service/module**.

◆ 1. User & Role Management Service

- User creation
- Role assignment
- Permission control

◆ 2. Job Management Service

- Job requisition
- Job approval workflow
- Job publishing

◆ 3. Candidate Management Service

- Candidate profile
- Resume handling
- Status lifecycle

- Duplicate detection

◆ **4. Interview Management Service**

- Interview scheduling
- Feedback collection

◆ **5. Offer Management Service**

- Offer generation
- Offer status tracking
- Compensation details

◆ **6. Workflow & Approval Service**

- Multi-level approvals
- Blueprint stage control
- Assignment rules

◆ **7. Vendor Management Service**

- Vendor onboarding
- Candidate submission
- Vendor performance tracking

◆ **8. Referral Management Service**

- Referral submission
- Referral tracking

◆ **9. Resume Parsing Service**

- Resume upload
- Field extraction
- Auto candidate creation

◆ 10. Notification Service

- Email
- SMS
- In-app notifications

◆ 11. Report & Analytics Service

- KPI calculation
- Custom report generation
- Dashboard metrics

◆ 12. Career Site & Portal Service

- Public job listing
 - Candidate application
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2.4 Integration Layer

External Integrations:

- Email servers (SMTP / Gmail / Outlook)
 - E-signature (Zoho Sign / DocuSign)
 - Job boards
 - Background verification APIs
 - Calendar (Google / Outlook)
 - Telephony APIs
 - Slack / Teams
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2.5 Database Layer

Recommended:

- PostgreSQL / MySQL

Core Tables:

User Management

- Users
- Roles
- Permissions

Recruitment Core

- Job_Openings
- Candidates
- Applications
- Interviews
- Offers
- Feedback

Vendor & Referral

- Vendors
- Vendor_Submissions
- Referrals

Workflow

- Approval_Processes
- Blueprint_Stages
- Tasks

Reports

- KPI_Metrics
- Activity_Logs

2.6 File Storage Layer

Used for:

- Resumes
- Offer letters
- Attachments
- Interview documents

Recommended:

- AWS S3 / Azure Blob / Local storage

3. Authentication & Security Architecture

Authentication:

- JWT / OAuth 2.0
- SSO (optional)

Authorization:

- Role-Based Access Control (RBAC)

Security Features:

- API encryption (HTTPS)
 - Audit logs
 - Data privacy controls
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4. Recruitment Workflow Architecture

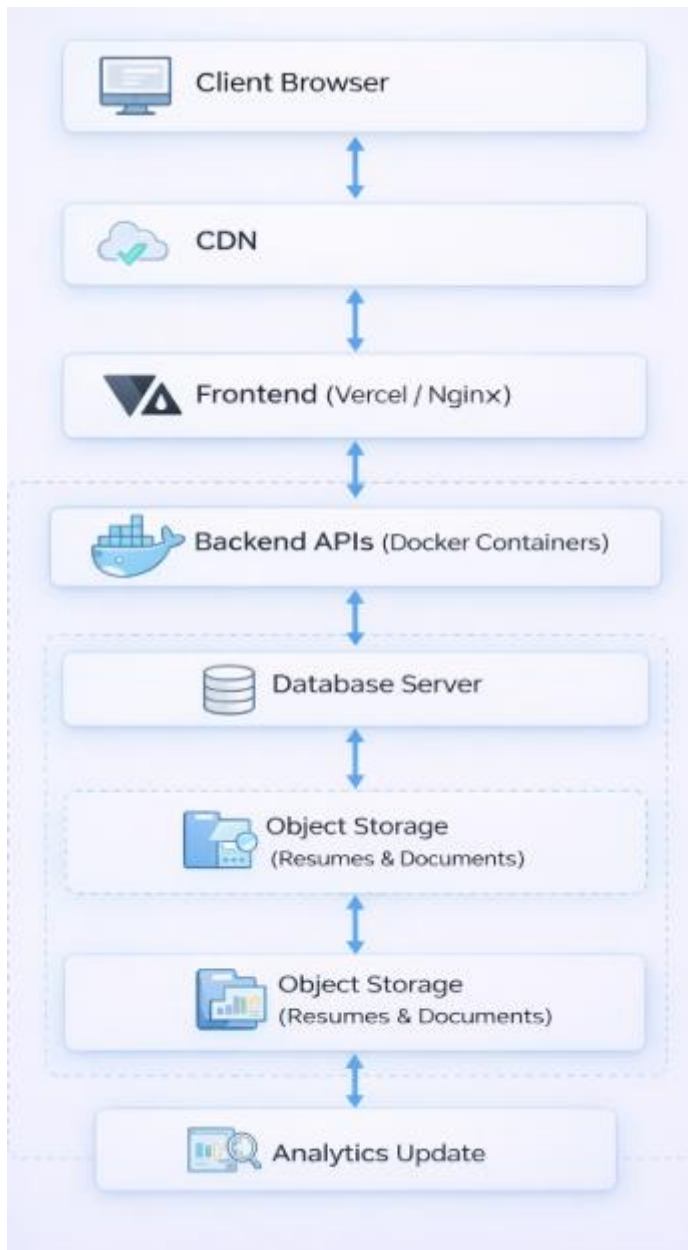


5. Dashboard & KPI Data Flow



6. Deployment Architecture (Recommended)

Cloud-Based:



7. Scalability Design

- Stateless backend services
- Microservices-ready modules
- Load balancer support
- Separate analytics processing

8. Optional AI Layer (Future Enhancement)

- Resume-job matching
- Candidate ranking
- Hiring prediction analytics
- Chatbot for candidate queries